

Transactions & Concurrency Control

1. Transactions and ACID Properties

A transaction is a single logical unit of work that accesses and possibly modifies the contents of a database.

To ensure integrity, transactions must satisfy the ACID properties:

- Atomicity: Either all operations of the transaction are reflected properly in the database, or none are.
- Consistency: Execution of a transaction in isolation preserves the consistency of the database.
- Isolation: Multiple transactions can execute concurrently without interfering with each other.
- Durability: After a transaction completes successfully, the changes it has made to the database persist, even if there are system failures.

2. Transaction States and Serializability

A transaction can be in one of the following states: Active, Partially Committed, Failed, Aborted, or Committed.

Serializability ensures that a concurrent schedule of transactions is equivalent to a serial schedule, guaranteeing consistency.

3. Concurrency Control and Locking

Concurrency control manages simultaneous operations without conflicts.

- Shared Lock (Read Lock): Allows a transaction to read but not update an item.
- Exclusive Lock (Write Lock): Allows a transaction to both read and update an item.

4. Deadlock and Two-Phase Locking

A deadlock occurs when two or more transactions are waiting indefinitely for one another to release locks. Two-Phase Locking (2PL) is a protocol that requires transactions to acquire all locks before releasing any (Growing phase followed by Shrinking phase).

5. Recovery Techniques

Database recovery restores the database to a consistent state after a failure. Techniques include log-based recovery (undo/redo logs) and shadow paging.